Data Structures

Subject: Computer Science | NoteShare Library

DATA STRUCTURES

1. ARRAYS
Access O(1), Search O(n)
Fixed size, contiguous

2. LINKED LISTS
Singly, Doubly, Circular
Access O(n), Insert head O(1)

3. STACKS (LIFO)
push, pop, peek O(1)
Apps: Function calls, Undo, DFS

4. QUEUES (FIFO)
enqueue, dequeue O(1)
Circular, Priority, Deque
Apps: BFS, Scheduling

5. TREES
Binary, BST, AVL, Red-Black
Traversals:
Inorder: L-Root-R
Preorder: Root-L-R
Postorder: L-R-Root
BST: O(log n) avg

6. HEAPS
Max/Min
Insert O(log n)
Extract O(log n)
Peek O(1)

7. GRAPHS
Matrix O(V), List O(V+E)
BFS, DFS O(V+E)

8. HASH TABLES
O(1) average
Chaining, Open addressing
Load factor n/m

9. SORTING
Bubble O(n)
Selection O(n)
Insertion O(n)
Merge O(nlogn)
Quick O(nlogn) avg
Heap O(nlogn)